

# Statistics: F-Test and One-Way ANOVA

## Comprehensive Lecture Notes with Exercises

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January 27, 2026

### Learning Objectives

By the end of this lecture, students will be able to:

1. Define the relationship between the F-test and ANOVA.
  2. Verify the four essential assumptions required for an ANOVA test.
  3. Calculate the F-statistic manually using the ratio of variances.
  4. Perform complete ANOVA calculations for one-way layout.
  5. Interpret ANOVA results and make statistical decisions.
  6. Apply post-hoc tests when ANOVA indicates significant differences.
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## 1 Conceptual Overview

The **ANOVA** (Analysis of Variance) allows us to compare the means of three or more groups simultaneously. It relies on the **F-Test**, which is a ratio of variances.

## 1.1 Key Terms and Formulas

**Definition 1** (Factor). *The independent variable being manipulated in an experiment (e.g., Medicine Type, Teaching Method).*

**Definition 2** (Level). *The different categories or treatments within a factor (e.g., 5mg, 10mg, 15mg of a drug).*

**Definition 3** (F-Statistic). *The ratio of between-group variance to within-group variance:*

$$F = \frac{MS_{between}}{MS_{within}} = \frac{s_{between}^2}{s_{within}^2}$$

where:

- $MS_{between}$  = Mean Square Between Groups (explained variance)
- $MS_{within}$  = Mean Square Within Groups (unexplained variance)

## 2 Assumptions in ANOVA

Before performing an ANOVA test, the following conditions must be met:

1. **Normality:** The residuals (errors) should be normally distributed.
2. **Absence of Outliers:** Data should be cleaned of extreme values.
3. **Homogeneity of Variance (Homoscedasticity):**

$$\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \dots = \sigma_k^2$$

4. **Independence:** Observations must be independent of each other.

## 3 Mathematical Foundation

### 3.1 ANOVA Notation

- $k$  = number of groups/treatments
- $n_i$  = number of observations in group  $i$
- $N = \sum_{i=1}^k n_i$  = total number of observations
- $x_{ij}$  =  $j$ th observation in group  $i$
- $\bar{x}_i$  = mean of group  $i$
- $\bar{\bar{x}}$  = grand mean (overall mean)

### 3.2 Sum of Squares Calculations

$$SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{\bar{x}})^2 \quad (\text{Total Sum of Squares})$$

$$SSB = \sum_{i=1}^k n_i (\bar{x}_i - \bar{\bar{x}})^2 \quad (\text{Between Groups Sum of Squares})$$

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 \quad (\text{Within Groups Sum of Squares})$$

$$SST = SSB + SSW$$

### 3.3 Mean Squares and F-Statistic

$$MS_{between} = \frac{SSB}{df_{between}} = \frac{SSB}{k-1}$$

$$MS_{within} = \frac{SSW}{df_{within}} = \frac{SSW}{N-k}$$

$$F = \frac{MS_{between}}{MS_{within}} \sim F(k-1, N-k)$$

## 4 Hypothesis Testing

- **Null Hypothesis ( $H_0$ ):** All group means are equal.

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

- **Alternative Hypothesis ( $H_1$ ):** At least one group mean is different.

$$H_1 : \exists i, j \text{ such that } \mu_i \neq \mu_j$$

## 5 Worked Examples

### 5.1 Example 1: Basic F-Test for Variance Comparison

**Example 1.** Compare the variances of two samples: Sample 1 ( $s_1^2 = 100, n_1 = 61$ ) and Sample 2 ( $s_2^2 = 70, n_2 = 21$ ) at  $\alpha = 0.05$ .

**Solution:**

1. **Hypotheses:**

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

2. **Calculate F-statistic:** Always place larger variance in numerator.

$$F_{calc} = \frac{s_{max}^2}{s_{min}^2} = \frac{100}{70} = 1.4286$$

3. **Determine Degrees of Freedom:**

$$df_1 = n_1 - 1 = 61 - 1 = 60, \quad df_2 = n_2 - 1 = 21 - 1 = 20$$

4. **Critical Value:** For  $\alpha = 0.05$ ,  $df_1 = 60$ ,  $df_2 = 20$ ,

$$F_{crit} = F_{0.025}(60, 20) \approx 2.223$$

5. **Decision:** Since  $F_{calc}(1.429) < F_{crit}(2.223)$ , we fail to reject  $H_0$ .

6. **Conclusion:** There is insufficient evidence to conclude that the variances are different at the 0.05 significance level.

## 5.2 Example 2: Complete One-Way ANOVA Calculation

**Example 2.** A researcher tests three different teaching methods on student performance. The test scores are:

- Method A: 85, 88, 92, 87, 90 ( $n=5$ )
- Method B: 78, 82, 80, 85, 79, 83 ( $n=6$ )
- Method C: 91, 89, 93, 95, 90, 92, 94 ( $n=7$ )

Test if there's a significant difference between methods at  $\alpha = 0.05$ .

**Solution:**

1. Calculate group means:

$$\bar{x}_A = \frac{85 + 88 + 92 + 87 + 90}{5} = 88.4$$

$$\bar{x}_B = \frac{78 + 82 + 80 + 85 + 79 + 83}{6} = 81.167$$

$$\bar{x}_C = \frac{91 + 89 + 93 + 95 + 90 + 92 + 94}{7} = 92$$

2. Calculate grand mean:

$$\bar{\bar{x}} = \frac{5 \times 88.4 + 6 \times 81.167 + 7 \times 92}{5 + 6 + 7} = \frac{442 + 487 + 644}{18} = \frac{1573}{18} = 87.389$$

3. Calculate Sum of Squares:

$$\begin{aligned}SSB &= 5(88.4 - 87.389)^2 + 6(81.167 - 87.389)^2 + 7(92 - 87.389)^2 \\&= 5(1.011)^2 + 6(-6.222)^2 + 7(4.611)^2 \\&= 5.111 + 232.296 + 148.889 = 386.296\end{aligned}$$

$$SSW_A = (85 - 88.4)^2 + (88 - 88.4)^2 + (92 - 88.4)^2 + (87 - 88.4)^2 + (90 - 88.4)^2 = 33.2$$

$$SSW_B = (78 - 81.167)^2 + \cdots + (83 - 81.167)^2 = 45.667$$

$$SSW_C = (91 - 92)^2 + \cdots + (94 - 92)^2 = 30.0$$

$$SSW = 33.2 + 45.667 + 30.0 = 108.867$$

$$SST = SSB + SSW = 386.296 + 108.867 = 495.163$$

4. Calculate Mean Squares:

$$MSB = \frac{SSB}{k - 1} = \frac{386.296}{2} = 193.148$$

$$MSW = \frac{SSW}{N - k} = \frac{108.867}{15} = 7.258$$

5. Calculate F-statistic:

$$F = \frac{MSB}{MSW} = \frac{193.148}{7.258} = 26.61$$

6. Determine critical value:

$$F_{crit} = F_{0.05}(2, 15) = 3.682$$

7. Decision: Since  $F_{calc}(26.61) > F_{crit}(3.682)$ , reject  $H_0$ .

8. Conclusion: There is a significant difference in test scores among the three teaching methods at the 0.05 level.

## 6 Exercise Problems

### 6.1 Basic F-Test Problems

1. **Variance Comparison:** Two machines produce components. Machine A has variance  $s_A^2 = 0.025$  (n=25), Machine B has variance  $s_B^2 = 0.018$  (n=20). Test if the variances are different at  $\alpha = 0.01$ .

2. **F-Statistic Calculation:** Calculate the F-statistic for comparing variances when:

(a)  $s_1^2 = 150, n_1 = 31$  and  $s_2^2 = 90, n_2 = 21$

(b)  $s_1^2 = 0.45, n_1 = 16$  and  $s_2^2 = 0.38, n_2 = 16$

(c)  $s_1^2 = 12.5, n_1 = 10$  and  $s_2^2 = 8.2, n_2 = 15$

## 6.2 ANOVA Calculation Problems

3. **Simple ANOVA:** Test the following data for differences between groups at  $\alpha = 0.05$ :
- Group 1: 10, 12, 14, 11, 13
  - Group 2: 8, 9, 10, 11, 7
  - Group 3: 15, 16, 14, 17, 16, 15
4. **ANOVA with Equal Sample Sizes:** Four teaching methods were tested with 5 students each:
- Method 1: 75, 78, 80, 72, 76
  - Method 2: 85, 88, 82, 87, 84
  - Method 3: 90, 92, 88, 91, 89
  - Method 4: 70, 72, 75, 71, 73

Perform ANOVA at  $\alpha = 0.01$ .

5. **ANOVA Table Completion:** Complete the following ANOVA table and test at  $\alpha = 0.05$ :

| Source  | SS    | df | MS | F |
|---------|-------|----|----|---|
| Between | 245.6 | 3  | ?  | ? |
| Within  | 156.8 | 16 | ?  |   |
| Total   | ?     | 19 |    |   |

## 6.3 Conceptual Problems

6. **Assumption Checking:** Describe how you would check each of the four ANOVA assumptions for a given dataset.
7. **Degrees of Freedom:** For an experiment with 4 treatments and total sample size of 40, calculate:
- (a)  $df_{between}$
  - (b)  $df_{within}$
  - (c)  $df_{total}$
8. **Hypothesis Formulation:** Write the null and alternative hypotheses for:
- (a) Comparing 5 different fertilizers on plant growth
  - (b) Testing 3 advertising strategies on sales
  - (c) Evaluating 4 drug dosages on pain relief

## 6.4 Advanced Problems

9. **Post-Hoc Analysis:** After finding a significant F-test ( $F=8.24$ ,  $p<0.01$ ) for 4 groups with  $n=10$  each, which post-hoc test would you use and why?
10. **Power Calculation:** For a one-way ANOVA with  $k=3$ ,  $n=20$  per group,  $\alpha = 0.05$ , and expected effect size  $f=0.4$ , calculate the approximate power of the test.
11. **Violation of Assumptions:** What would you do if:
  - (a) The normality assumption is violated?
  - (b) The homogeneity of variance assumption is violated?
  - (c) There are outliers in the data?

## 7 Challenge Problems

1. **Unequal Variances ANOVA:** Given the following data with unequal variances, perform a Welch's ANOVA:
  - Group A:  $\bar{x}_1 = 50$ ,  $s_1^2 = 25$ ,  $n_1 = 10$
  - Group B:  $\bar{x}_2 = 55$ ,  $s_2^2 = 100$ ,  $n_2 = 15$
  - Group C:  $\bar{x}_3 = 48$ ,  $s_3^2 = 36$ ,  $n_3 = 12$
2. **ANOVA Derivation:** Prove that  $SST = SSB + SSW$  using algebraic manipulation.
3. **Sample Size Determination:** For a one-way ANOVA with  $k=4$  groups, desired power= $0.80$ ,  $\alpha = 0.05$ , and medium effect size ( $f=0.25$ ), determine the required sample size per group.

## Answers to Selected Problems

### Basic F-Test Solutions

1.  $F_{calc} = 0.025/0.018 = 1.389$ ,  $df_1 = 24$ ,  $df_2 = 19$ ,  $F_{crit}(0.005, 24, 19) \approx 3.72$ , Fail to reject  $H_0$ .
2. (a)  $F = 150/90 = 1.667$ ,  $df_1 = 30$ ,  $df_2 = 20$   
(b)  $F = 0.45/0.38 = 1.184$ ,  $df_1 = 15$ ,  $df_2 = 15$   
(c)  $F = 12.5/8.2 = 1.524$ ,  $df_1 = 9$ ,  $df_2 = 14$

## ANOVA Table Completion

| Source  | SS    | df | MS    | F    |
|---------|-------|----|-------|------|
| Between | 245.6 | 3  | 81.87 | 8.36 |
| Within  | 156.8 | 16 | 9.8   |      |
| Total   | 402.4 | 19 |       |      |

$F_{crit}(3, 16) = 3.24$ , Reject  $H_0$  since  $8.36 > 3.24$ .

## Degrees of Freedom

3. (a)  $df_{between} = k - 1 = 4 - 1 = 3$
- (b)  $df_{within} = N - k = 40 - 4 = 36$
- (c)  $df_{total} = N - 1 = 39$

### Important

**Note:** Complete solutions for all problems are available in the accompanying solution manual. For additional practice, work through each problem step-by-step, verifying assumptions at each stage of the analysis.

## Additional Resources

- [Penn State STAT 500: Applied Statistics](#)
- [Khan Academy: Statistics and Probability](#)
- [Real Statistics: One-Way ANOVA](#)